

Part II, Chapter 4: Simulated Agency: Consciousness as Sparse Projection

4.1 Introduction: Agency in an Entropic Universe

The Relativistic Scalar-Vector Plenum (RSVP) framework, as developed in Part I, reimagines computation and cognition as field-theoretic processes governed by scalar coherence (Φ), vectorial semantic flows ($\mathbf{v}$), and entropic dispersion (S). While Chapter 3 outlined a semantic infrastructure to replace rigid file-based systems, the question of agency—how cognitive entities navigate and act within this plenum—remains central to transcending the attention-extractive paradigms of contemporary technology. This chapter introduces the concept of simulated agency as a sparse projection mechanism, where consciousness emerges as a low-dimensional representation of high-dimensional semantic states, optimized to balance predictive accuracy and entropic flexibility.

Drawing on the Chain of Memory (CoM) paradigm and variational principles (Appendix B), we formalize agency as a dynamic process within $\mathcal{C}_{\text{RSVP}}$, the symmetric monoidal ∞ -category introduced in Chapter 3. By integrating sparse projection operators with the distributed architectures of HYDRA and TARTAN (Appendix D), we demonstrate how RSVP enables modular, decentralized agency that resists monopolistic cognitive capture.

4.2 The Chain of Memory: A Recursive Model of Agency

Agency, in the RSVP framework, is modeled as a recursive update process over a memory state M_t , which encapsulates the semantic history of an agent at time t . As defined in Appendix B.1, the memory state evolves according to:

$$M_{t+1} = \Pi(M_t, I_t; \theta),$$

where Π is a sparse projection operator, I_t represents external stimuli, and θ denotes parameters encoding learned or innate semantic attractors. The operator Π selects a low-dimensional subspace of the high-dimensional semantic manifold Ω , ensuring that agency operates efficiently within the constraints of cognitive bandwidth.

This recursive structure, termed the Chain of Memory (CoM), mirrors Bayesian inference loops but is grounded in the field dynamics of RSVP. Specifically, the scalar field Φ (Appendix A.1) defines the coherence potential guiding the projection, while the vector field $\mathbf{v}$ (Appendix A.2) directs the flow of semantic updates. The entropy field S (Appendix A.3) ensures that the projection respects informational uncertainty, preventing overfitting to local attractors—a common pathology in attention-driven systems.

4.3 Variational Principles for Simulated Agency

The optimization of agency within RSVP is formalized through a variational objective, analogous to a free-energy functional, as specified in Appendix B.2:

$$\mathcal{F}[M_t, \Phi, S] = \mathbb{E}_{M_t}[\ln P(I_t | M_t)] + \beta S(M_t),$$

where $\mathbb{E}_{M_t}[\ln P(I_t | M_t)]$ measures the predictive accuracy of the memory state given new stimuli, and $\beta S(M_t)$ penalizes excessive entropic rigidity. The parameter $\beta > 0$ balances

the trade-off between accuracy and flexibility, ensuring that agency remains adaptive without collapsing into deterministic paths.

Minimizing $\mathcal{F}$ corresponds to finding a sparse representation of the semantic state that aligns with the RSVP fields. The governing dynamics are:

$$\frac{\partial M_t}{\partial t} = -\nabla_{M_t} \mathcal{F} = -\nabla_{M_t} \mathbb{E}_{M_t} [\ln P(I_t | M_t)] - \beta \nabla_{M_t} S(M_t).$$

This gradient descent process is constrained by the PDEs of the RSVP plenum (Appendix A), particularly the entropy evolution equation:

$$\frac{\partial S}{\partial t} + \nabla \cdot (\mathbf{v}S) = \kappa \nabla^2 S + \Gamma(\Phi, \mathbf{v}),$$

ensuring that agency evolves in harmony with the plenum’s entropic structure.

4.4 Category-Theoretic Formalization of Agency

To formalize agency within $\mathcal{C}_{\text{RSVP}}$, we treat agents as functors from a category of temporal states $\mathcal{T}$ to $\mathcal{C}_{\text{RSVP}}$:

$$A : \mathcal{T} \rightarrow \mathcal{C}_{\text{RSVP}},$$

where $\mathcal{T}$ is a poset category of time steps $t \leq t'$, and $A(t) = M_t$ maps each time to a semantic module. Morphisms in $\mathcal{T}$ correspond to temporal transitions, and the functor A ensures that these transitions preserve the monoidal structure of $\mathcal{C}_{\text{RSVP}}$. The sparse projection Π is represented as a natural transformation:

$$\Pi : A \rightarrow A',$$

where A' is a lower-dimensional functor representing the projected agency state.

This categorical approach enables agents to operate within the fibered structure of $\mathcal{C}_{\text{RSVP}}$ (Appendix C.3), allowing cross-domain interactions via pullbacks. For instance, an agent integrating sensory data (I_t) from multiple domains forms a pushout in $\mathcal{C}_{\text{RSVP}}$:

$$\begin{array}{ccc} M_t & \xrightarrow{\Pi} & M_{t+1} \\ \downarrow \iota & & \downarrow \iota' \\ D_1 & \xrightarrow{f} & D_2 \end{array}$$

Here, $D_1, D_2 \in \mathcal{B}$ are contextual domains, and ι, ι' are embeddings, ensuring that agency remains coherent across knowledge fibers.

4.5 HYDRA and TARTAN: Distributed Agency Architectures

The practical realization of simulated agency leverages the HYDRA and TARTAN architectures (Appendix D). HYDRA models agents as nodes $N_i = (\Phi_i, \mathbf{v}_i, S_i)$, with local dynamics coupled via an adjacency matrix A :

$$\mathbf{v}_i(t+1) = \sum_j A_{ij} \mathbf{v}_j(t) + f_i(t).$$

This distributed structure ensures that each agent’s semantic flow aligns with global coherence, as dictated by Φ . TARTAN extends this by acting as a functor:

$$T : \mathcal{C}_{\text{RSVP}}^{\text{local}} \rightarrow \mathcal{C}_{\text{RSVP}}^{\text{global}},$$

mapping local agency modules to a globally coherent state, preserving sparse interactions and global invariants (Appendix D.2).

These architectures enable scalable, decentralized agency, where multiple agents coordinate without centralized control, countering the monopolistic tendencies of attention economies. The sparse projection operator Π is implemented via low-rank tensor factorization (Appendix G), ensuring computational efficiency even in high-dimensional semantic spaces.

4.6 Consciousness as Emergent Sparse Projection

Within RSVP, consciousness is not a singular entity but an emergent property of sparse projection across the plenum. The recursive update of M_t via Π generates a low-dimensional manifold of cognitive states, where the interplay of Φ , $\mathbf{v}$, and S produces coherent, adaptive behavior. This contrasts with attention-extractive systems, which collapse cognitive trajectories into predetermined attractors, as described in Chapter 1.

Formally, consciousness arises as the stable fixed point of the variational dynamics:

$$M^* = \arg \min_{M_t} \mathcal{F}[M_t, \Phi, S],$$

where M^* represents a cognitive state that optimally balances predictive accuracy and entropic flexibility. The categorical structure ensures that this fixed point is robust across domains, enabling agents to navigate complex semantic landscapes without loss of autonomy.

4.7 Conclusion: Toward Ethical Agency

By modeling agency as a sparse projection within the RSVP plenum, this chapter establishes a framework for consciousness that is modular, decentralized, and entropy-respecting. Unlike the coercive optimization of Big Tech platforms, RSVP-based agency empowers cognitive entities to explore semantic trajectories freely, guided by coherence potentials rather than external attractors. The next chapters will explore how these principles extend to entropic economics (Chapter 5) and ethical AI design (Chapter 6), leveraging the distributed architectures of HYDRA and TARTAN to build a cognitive commons.